



POLYNOMIALS

KEY POINTS



01

Polynomials

- An algebraic expressions of form $a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_1 x^1 + a_0$.
- $a_0, a_1, \dots, a_n$ are real numbers and all indices of variable x are non-negative integers.
- $a_0, a_1 x, \dots, a_n x^n$ are terms and $a_0, a_1, \dots, a_n$ are coefficients.
- For e.g. $p(x) = 6x^3 + 4x^2 + 3x + 1$

02

Degree of Polynomial

- Highest power of variable is called degree.
- For e.g. The degree of the polynomial $6x^4 + 2x^3 + 3$ is 4.

03

Types of Polynomials

ON BASIS OF TERMS	ON BASIS OF DEGREE
Monomial ($2x$)	Zero (0)
Binomial ($x^2 + 3x$)	Constant (3)
Trinomial ($3x^3 - x^2 + 3x$)	Linear ($2x + 1$)
Quadrinomial ($2x^4 + 3x^3 - x^2 + 3x$)	Quadratic ($x^2 - 3$)
	Cubic ($x^3 - 4x + 3$)
	Bi-Quadratic ($x^4 - x + 5$)

04

Zero or Root of polynomial

- $x = r$ is a zero of a polynomial $p(x)$, if $p(r) = 0$
- For e.g. Check whether 2 and 3 are zero of given polynomial $p(x) = x^2 - 5x + 6$ or not.
Sol. $p(2) = (2)^2 - 5 \times 2 + 6 = 4 - 10 + 6 = 0$,
 $p(3) = (3)^2 - 5 \times 3 + 6 = 9 - 15 + 6 = 0$
 $\therefore$ 2 and 3 are the zeros of the polynomial $p(x)$.



05 Value of Polynomial

The value of a polynomial $f(x)$ at $x = a$ is obtained by substituting $x = a$ in given polynomial.

- **For e.g.** Find the value of polynomial $5x - 4x^2 + 3$ at $x = -1$.

Sol. $p(x) = 5x - 4x^2 + 3$

To find value of polynomial at $x = -1$ or $p(-1)$

So we put $x = -1$ in polynomial,

$$p(-1) = 5 \times (-1) - 4 \times (-1)^2 + 3$$

$$p(-1) = -5 - 4 + 3$$

$$p(-1) = -6$$

So value of polynomial at $x = -1$ is -6 .

06 Remainder Theorem

- If $p(x)$ is divided by $(x - a)$ then remainder is given by $p(a)$.
- **For e.g.** Find the remainder when polynomial $p(x) = x^3 - 4x^2 - 7x + 10$ is divided by $x - 2$.

Sol. $p(x) = x^3 - 4x^2 - 7x + 10$

$$p(2) = 2^3 - 4(2)^2 - 7(2) + 10$$

$$= 8 - 16 - 14 + 10 = -12$$

07 Factor Theorem

- If $(x - a)$ is a factor of $p(x)$, then $p(a) = 0$ and vice versa.
- **For e.g.** Examine whether $x + 2$ is a factor of $x^3 + 3x^2 + 5x + 6$.
Sol. We know that the zero of the polynomial $(x + 2)$ is -2 .
Let $p(x) = x^3 + 3x^2 + 5x + 6$
Then, $p(-2) = (-2)^3 + 3(-2)^2 + 5(-2) + 6 = -8 + 12 - 10 + 6 = 0$
Hence $x + 2$ is factor of given polynomial.
- According to the factor theorem, if $p(a) = 0$, then $(x - a)$ is a factor of $p(x)$.

08 Factorisation of Polynomials

- By taking common factor

For e.g. Factorise $12x^2 + 18x$.

Sol. $12x^2 + 18x$.

$$6x(2x + 3)$$

- By making perfect square

For e.g. Factorise $4x^2 + 9y^2 + 12xy$.

Sol. First we arrange the given expression $4x^2 + 9y^2 + 12xy$ in the form of $a^2 + 2ab + b^2$.

$$4x^2 + 12xy + 9y^2$$

$$= (2x)^2 + 2(2x)(3y) + (3y)^2$$

Now applying the formula of $a^2 + 2ab + b^2 = (a + b)^2$ then we get,

$$= (2x + 3y)^2$$

$$= (2x + 3y)(2x + 3y)$$

- By factor theorem

For e.g. Using factor theorem, factorize the polynomial $x^3 - 6x^2 + 11x - 6$.

Sol. Let $f(x) = x^3 - 6x^2 + 11x - 6$

The constant term is -6 and factors of -6 are $\pm 1, \pm 2, \pm 3, \pm 6$.

Putting $x = 1$ in $f(x)$, we have

$$f(1) = 1^3 - 6 \times 1^2 + 11 \times 1 - 6$$

$$= 1 - 6 + 11 - 6 = 0$$

Hence $(x - 1)$ is a factor of $f(x)$.

Similarly, $x - 2$ and $x - 3$ are factors of $f(x)$.

- By splitting the middle term

For e.g. Factorise $x^2 + 5x + 6$

Sol. $x^2 + 5x + 6$

$$x^2 + 3x + 2x + 6$$

$$x(x + 3) + 2(x + 3)$$

$$(x + 3)(x + 2)$$

09 Factorisation of cubic polynomials

- For e.g. Factorize the cubic polynomial $f(x) = x^3 - 5x^2 + 4x - 20$.

Sol. To factorize the polynomial $f(x)$, we will divide it into groups.

$$\begin{aligned}f(x) &= x^3 - 5x^2 + 4x - 20 \\&= (x^3 - 5x^2) + (4x - 20) \\&= x^2(x - 5) + 4(x - 5) && \text{[Taking common terms out]} \\&= (x - 5)(x^2 + 4)\end{aligned}$$

Now, since $x^2 + 4$ does not have real roots, so we have expressed the given cubic polynomial as a product of its factors $(x - 5)$ and $(x^2 + 4)$.

10 Algebraic Identities

- $(a + b)^2 = a^2 + 2ab + b^2$
 $(a - b)^2 = a^2 - 2ab + b^2$
 $a^2 - b^2 = (a + b)(a - b)$
 $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$
 $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
 $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
 $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
 $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
 $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$
- **Special case :** If $a + b + c = 0$ then $a^3 + b^3 + c^3 = 3abc$